Cross-asset dual momentum

Raising assets for a transparent monthly ETF portfolio

Research-stage B3: risk cap, explicit cash, no leverage

CAGR 6.89% | daily max drawdown -17.91%

Illustrative USD 10 million mandate; execution validation starts with a USD 1 million scenario

Momentum strategy landscape

Buy-and-hold: continuous market exposure

Relative momentum: rank assets against peers

Absolute momentum: compare an asset with its cash hurdle

Our dual design: three selected ETF slots, failed slots to cash

Futures CTAs can short declining markets; this portfolio cannot

Economic mechanism and contribution

Published hypotheses: delayed information, feedback and risk premia

Equity factor crashes: past losers rebound after panic states

ETF long/cash risks differ from short-loser equity momentum

Contribution: controlled comparisons with reproducible timing and accounting

Exact monthly holdings rule

Universe: SPY EFA EEM IEF TLT GLD DBC VNQ

Rank 12-month total returns; choose top 3

Each 1/3 slot invests only if its return exceeds matched cash

Example signals 12%, 8%, 1%; cash 2% -> two 1/3 ETFs +1/3 cash

Vol 16% -> q 0.625 ->20.83%, 20.83%, 58.33% cash

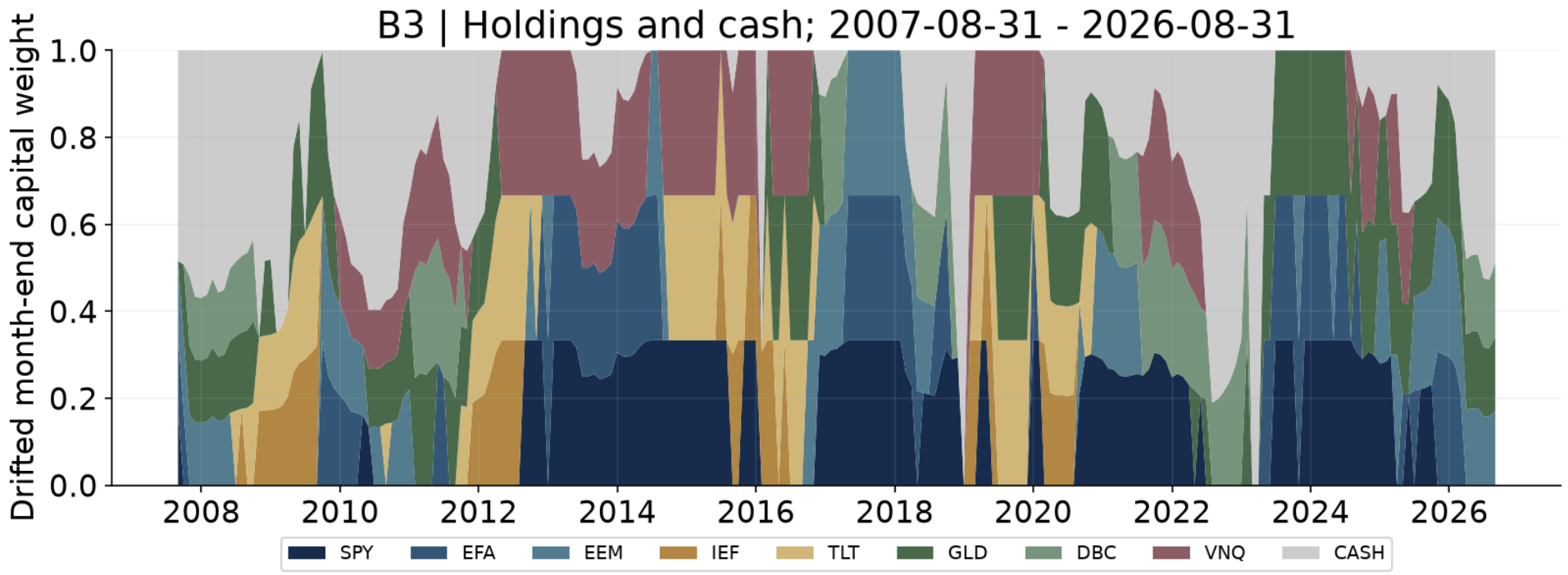
Risk cap and opportunity cost

B3 estimates 126-day gross B2 excess volatility

q=min(1, 10%/max(volatility, 5%)); remainder cash

Average cash 22.82% versus B2 5.22%

Protection reduces upside and does not guarantee maximum loss



Data, timing and historical validation

Real French, Cboe and Yahoo/yfinance snapshots

September cutoff; French RF/Mom available only through August

Signal penultimate close -> execute final close -> next-session return

All 8 ETF history begins 2006; B3 warmup -> September 2007 returns

Development to 2013 | validation 2014-19 | historical holdout 2020-26

Modeled net results versus benchmarks

B0: CAGR 6.40% | Sharpe 0.487 | max DD -37.92%

B1: CAGR 8.66% | Sharpe 0.607 | max DD -25.12%

B2: CAGR 7.95% | Sharpe 0.570 | max DD -25.15%

B3: CAGR 6.89% | Sharpe 0.614 | max DD -17.91%

60_40: CAGR 8.16% | Sharpe 0.719 | max DD -32.62%

SPY: CAGR 11.08% | Sharpe 0.662 | max DD -55.19%

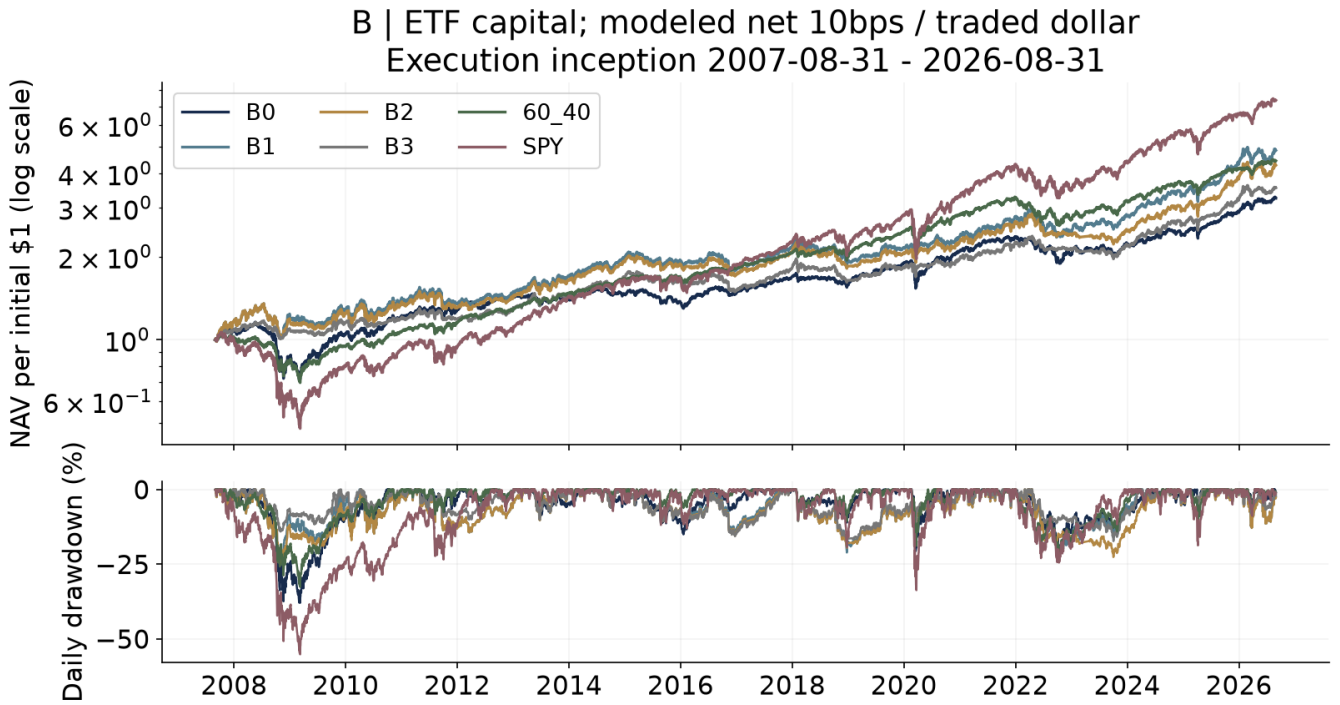
Lower drawdown comes with lower return

Relative selection improves historical return over equal weighting

Absolute filter scarcely improves worst drawdown here

Volatility cap provides the strongest ETF drawdown reduction

B3 trails SPY and 60/40 in full-history CAGR



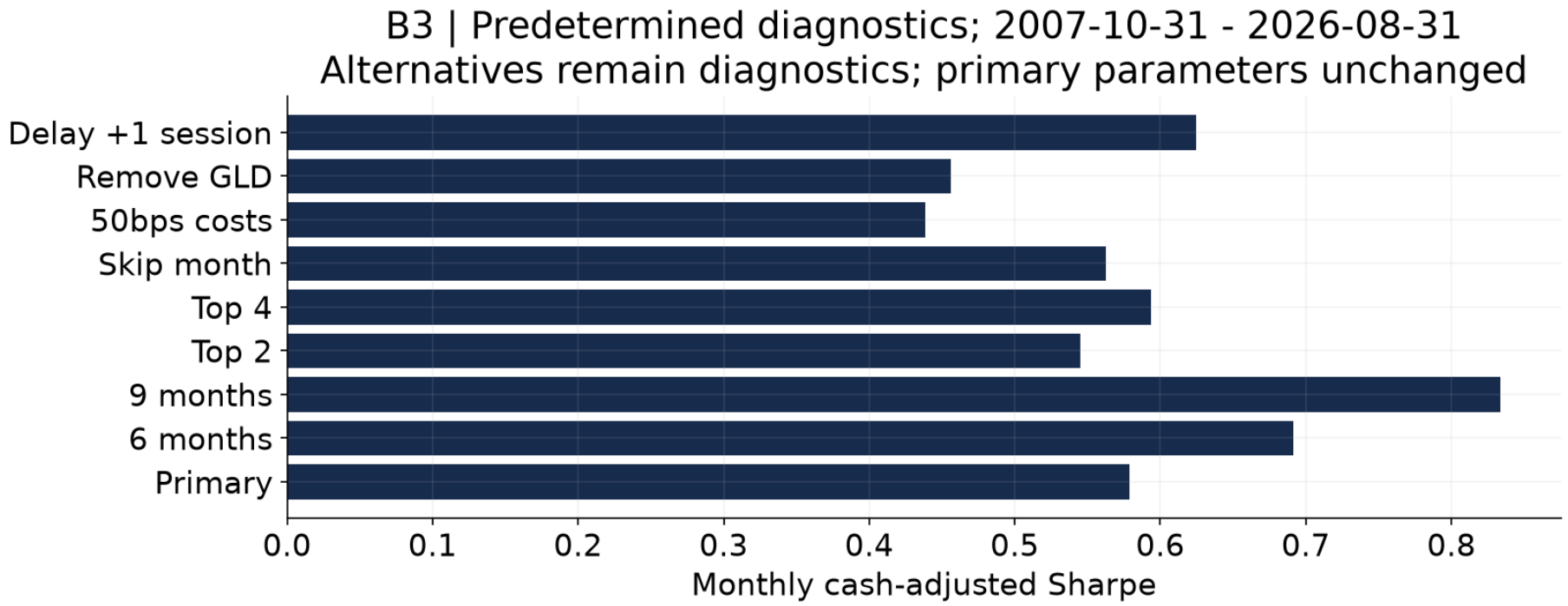
Robustness and uncertainty

Historical holdout B3: CAGR 10.46% | Sharpe 0.805

B3-minus 60/40 Sharpe 95% CI: [-0.598, 0.279]

Skip-month diagnostic worsens drawdown; gold exclusion hurts

All 84 ledger rows retained; stronger 9-month result is not adopted



Costs, liquidity and mandate size

B3 turnover 3.15x NAV/year; assumed 10 bps per dollar

1m/10m/50m AUM scenarios evolve with net NAV

10m B3 maximum DBC participation 19.17% of lagged ADV20

All-day vendor ADV does not certify closing-auction capacity

Validate 1m scenario first;10m single-close fills are not established

Evidence-supported advantages and risks

Advantages: explicit holdings, fixed rules, no leverage, cash audit

B3 reduced losses in 2022; it missed upside in 2020

Risks: whipsaw, reversals, gold dependence and duration overlap

Higher full-sample Sharpe than 60/40 is not demonstrated

Proposed research mandate and investor Q&A

Proposal: B3 research-stage risk-limited ETF design

Why more cash? It buys lower exposure and sacrifices return

Why not B1? Keep it as the higher-return comparator

Next milestone: independent data, auction costs, prospective paper signals

Feed failure: preserve holdings, verify data, escalate and log

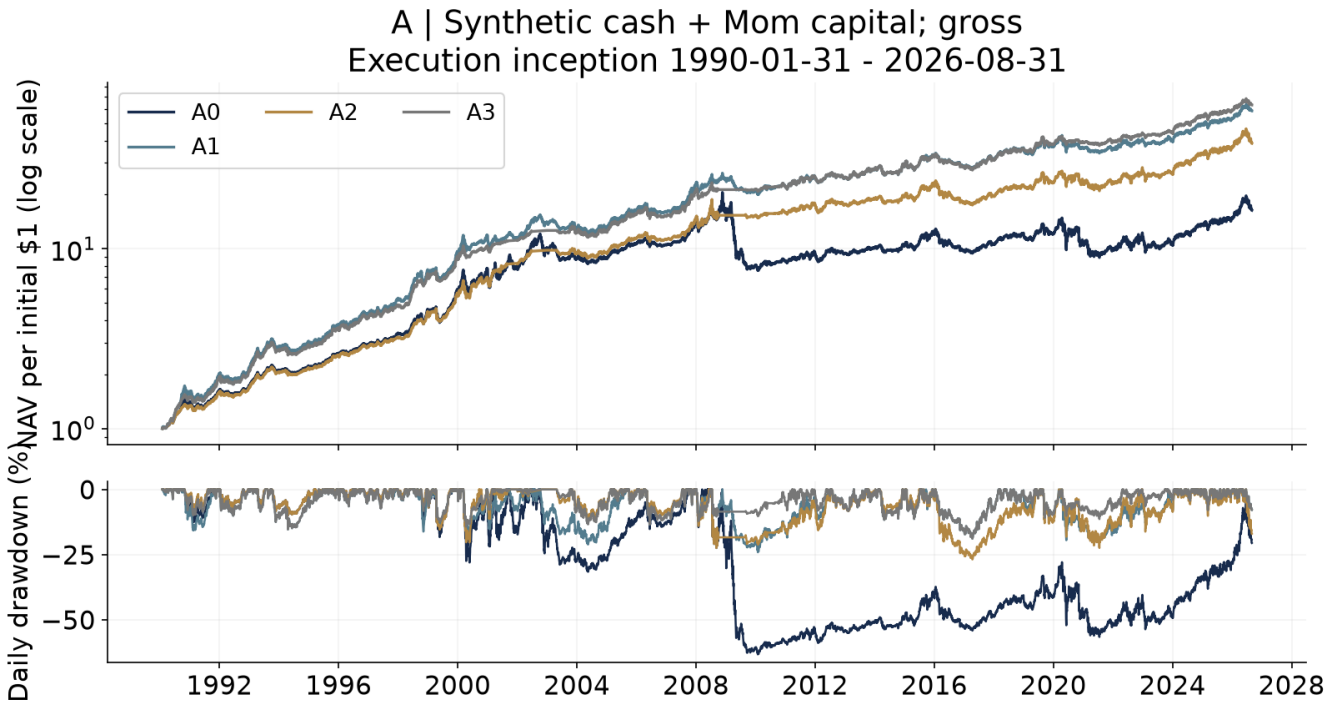
Appendix | Factor experiment

A0 gross synthetic CAGR 7.94% | drawdown -63.29%

A3 gross synthetic CAGR 11.99% | drawdown -18.89%

Bear 24 months + VIX>25 -> exposure 0 when both flags

RF + a*Mom: funding/borrow/internal turnover not measured



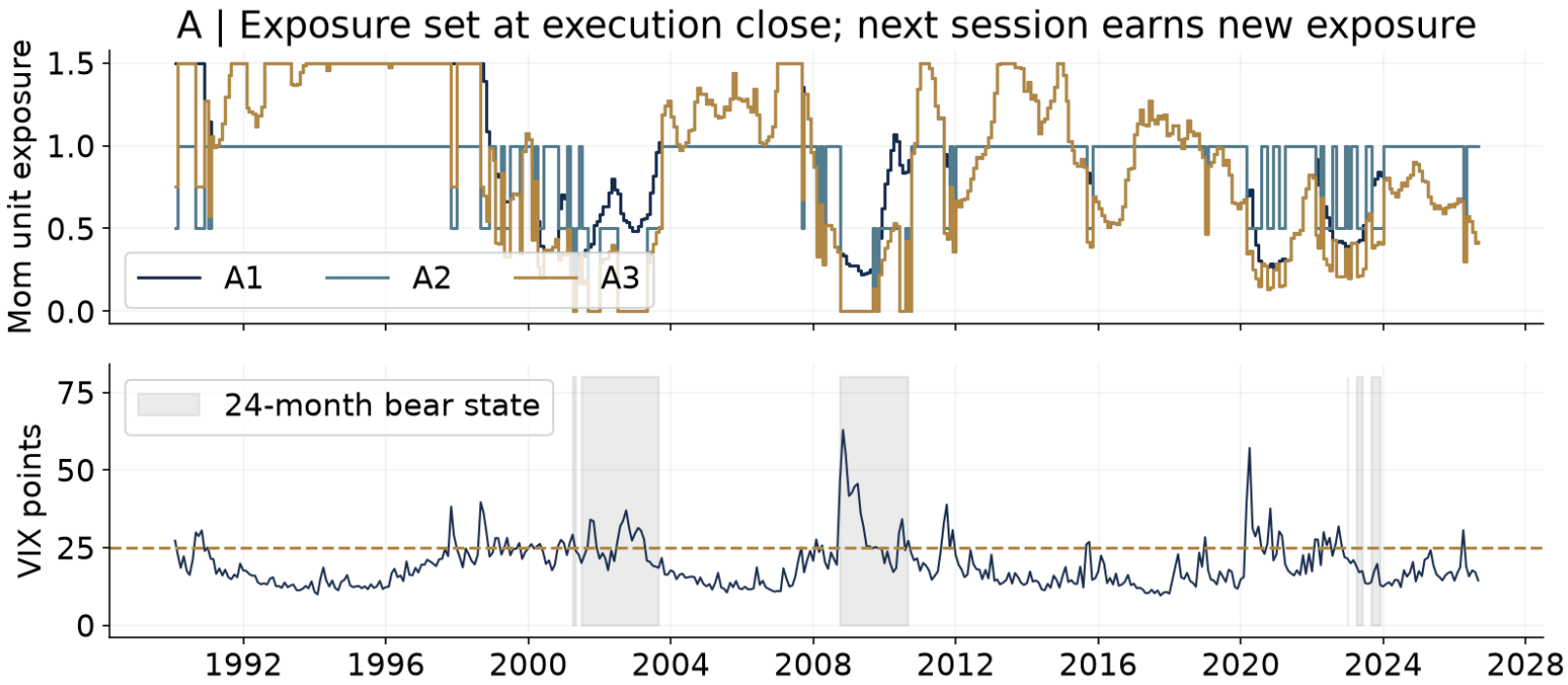
Appendix | Factor exposures and panic states

A0=1 | A1=vol scale | A2=state scale | A3=combined

a_vol=min(1.5, 10%/max(sigma 126, 5%))

Market total return compounds Mkt-RF + RF

Published loser-rebound mechanism motivates the fixed heuristic



Appendix | Accounting and statistical conventions

Daily weights drift; rebalance only at execution close

T=sum risky |target-pretrade|; full A-to-B switch T=2

NAV_after=NAV_before*(1+gross)*(1-0.001*T)

Sharpe=sqrt(12)*mean(monthly excess)/sample_sd

CAGR ACT/365.25 | HAC3 | downside RMS Sortino

Appendix | Exact component evidence

Decimals to five places; ETF modeled net, factor synthetic gross

A0: CAGR 0.07938; Sharpe 0.39282; DD -0.63288

A1: CAGR 0.11762; Sharpe 0.75071; DD -0.24129

A2: CAGR 0.10500; Sharpe 0.70178; DD -0.26873

A3: CAGR 0.11994; Sharpe 0.89354; DD -0.18893

B0: CAGR 0.06405; Sharpe 0.48708; DD -0.37924

B1: CAGR 0.08658; Sharpe 0.60665; DD -0.25123

B2: CAGR 0.07953; Sharpe 0.57005; DD -0.25147

B3: CAGR 0.06891; Sharpe 0.61357; DD -0.17911

Appendix | Sources and reproduction

Course lecture Momentum; Homework 2 handout

Daniel-Moskowitz 2016; Asness et al. 2014

Kolanovic-Wei 2015; Baltas-Kosowski supplied 2013 version

French official library | Cboe VIX | Yahoo via yfinance 1.7.0

README: cached run, tests, reports and executed notebooks